

# Goals, Prerequisites, Target Audience

**T**his book has at least two goals. With a modest body of work that you are holding in your hands we want to, first, introduce our readers to the technical essence of inversive geometry in an elementary fashion and open up the horizon for a consequent deeper study of this layer of mathematics.

We will do so by introducing the concept of inversion of the real plane in a circle by using only 1) the means of synthetic, read *straightedge and compass*, Euclidean geometry and 2) the minimal amount of preexisting mathematical knowledge.

That is, in this book we will not use the machinery of Analytic Geometry and we will not use the notion of the polar of a point with respect to a quadratic curve.

Second, using the notion of inversion of the plane in a circle, we want to introduce our readers to what Mathematics really is. By not using any symbolic shuffling on the one hand and by doing the work that is Mathematics on the other hand, we will bring to light the fact that:

- Mathematics is an art
- Mathematics is an art of telling a story
- Mathematics is an art of telling a story of how and why

## Front Matter

- Mathematics is an art of telling a story of how and why wrapped into the technical language of IF P THEN Q implications, logical connectives NOT, AND, OR and the universal, for all, and the existential, there exists, quantifiers

In addition, we will see that Mathematics is a process and that in Mathematics the first class citizen is her majesty *idea* for Mathematical proverbial needle is moved by generation and re-generation and combination and re-combination of the said ideas, not symbols.

## Prerequisites

The prerequisites for a successful journey through this elementary introduction to inversive geometry are quite low.

The reader is only expected to 1) possess the basic situational awareness and 2) be familiar with the elementary Euclidean geometry at the level of an average upper middle or lower high school reasonably well.

Above all, however, it is the genuine desire to learn something new and the willingness to step out of one's comfort zone are the valued traits that will ultimately determine the level of success of undertaking this journey and engaging the subject matter.

In order to grow mathematically as time goes by, the reader is also expected to rise to the challenge of following and writing mathematical proofs for our discourse will be proofs-based. Do your best to solve the suggested exercises and problems on your own before looking our solutions up.

## Target Audience

From the goals of this book and its prerequisites it follows that there is no age restriction to reading this book and, on the practical side, anyone between 12 and 112 years of age who is interesting in learning a beautiful mathematical concept is more than welcome to study this material.

## Structure

This book is organized into three big chunks which, for best results, should be consumed in this particular order:

- inversion in a circle in the real plane
- pencils of circles
- inversion in a circle in the complex plane

I. Inversion in a circle in the real plane is the core material onto which the rest of the relevant ideas are strung.

## Front Matter

We will develop that material by studying how inversion of the plane in a circle acts on:

- points
- straight lines
- circles
- angles and
- tangency

in that order.

Once these basics are covered, we will, next, solve a number of elegant problems that showcase the essence and the power of inversive geometry.

**II.** Inching beyond the basics, we will put the new knowledge to a test by developing the basic theory of pencils of circles.

**III.** Advancing the level of our knowledge in a different direction, we will look at the same ideas through the lens of a new tool, complex numbers.